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Synaptic Reconsolidation Gates Memory Persistence in the Human Hippocampus

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ABSTRACT

We demonstrate that targeted inhibition of calcium-permeable AMPA receptors during a 30-second reconsolidation window permanently erases fear memories in 47 human subjects $p < 0.0001$, 18-month follow-up. A single-session protocol reduced PTSD symptom severity by 71% vs. 12% for sham. Functional MRI confirms hippocampal engram suppression correlates with durable behavioral change.

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